

Quality control and compactness are the other aspects where the industry needs to improve upon. There has been lots of talk about open architecture and enterprise-wide solutions (sensing to MIS reporting), but it hasn't taken off to that extent where it would really make a difference to the end users.

Manufacturing convergence: Enabling plant-wide optimisation

Manufacturing convergence is the merger of traditionally disparate functions and systems across the enterprise. Through this union, all information sources can be streamlined to allow access to the right information, in the right place, at the right time and in the right format. This eases configuration, visualisation, maintenance and optimisation of processes and critical plant assets.

Key market drivers of manufacturing convergence. It is the merging of manufacturing and production systems—manufacturing convergence—with the entire corporate enterprise that allows manufacturers to constantly balance productivity, innovation, globalisation and sustainability.

Productivity. An era that once stressed 'maximum productivity' has evolved to 'optimum productivity' where machines, processes and factories dynamically adjust to meet continuous changes in demand. By tapping into an advanced process control solution that merges software with control technology, plants can optimise productivity—increasing production and throughput while reducing energy consumption.

Innovation. By using the automation technology available today, an entire plant can be modeled digitally so new

Some players and their offerings

GE has developed a whole range of wireless products like wireless I/Os, wireless sensors and modems, and single-board computers with latest technologies like VPX, VXS and CPCL. Smart Grid is a technology initiative driven by GE to cut down on power distribution and utilisation losses.

Rockwell Automation offers combustion control, condition monitoring, drive system, information solutions, integrated architecture, intelligent motor control, machine builder, maintenance, migration, on-machine power and energy management, process solution, safety and security, sustainable production and visualisation solutions.

Beginning with the design and manufacture of variable-speed AC motor drives in 1995, Delta today offers a wide range of products for industrial applications. Its products include AC motor drives, servo drives and motors, temperature controllers, programmable logic controllers, human-machine interface modules, optical encoders, communication converter modules, and CTA timers/counters/tachometers.

ConnectM's solutions are powered by cutting-edge M2M technology. These solutions leverage innovation in hardware design, embedded software, Web-based application and system integration to provide a good value proposition to customers. Remote asset management (telecom towers), automated meter reading, energy technology solutions, fleet management solution, and remote condition monitoring and predictive maintenance are some of the offerings.

production concepts can be tested virtually.

Reconfigurable and scalable manufacturing equipment can be quickly assembled to respond to fast-changing consumer preferences. Intelligent machines can monitor their own health, throughput and yield, and report corrective action to the maintenance staff. These innovations are making organisations more nimble and flexible in order to move new products to the market as requirements change.

Arkema, an industrial chemical producer, helped drive new product innovation and improved productivity with a new, fully integrated process automation system. Working with Rockwell Automation, the company merged control technologies, integrated HMI visualisation software into each application, added batch-management software to control recipes, and linked the system together via a communications interface to provide insight for third-party access.

Globalisation. Manufacturing convergence helps to provide organisations with systems that allow them to capitalise on changing conditions or, if necessary, reduce their risk.

Sustainability. Manufacturers are widening their focus from 'output efficiency' (how fast a product was made) to also include 'input efficiency' (what it took to make the product). The ability to reduce energy costs, measure and manage carbon-footprint and process emissions, as well as improving worker and plant safety and product integrity are all important aspects of a sustainability programme. This desire to track all aspects of manufacturing—from suppliers through production to end delivery—requires a heightened level of integration between manufacturing and other areas of the business.

Role of various technologies

The impact of industrial wireless, embedded intelligence and machine-to-machine (M2M) communications, Web services and applications integration for real-time information, and robots on industrial automation has been revolutionary.

Industrial wireless. Wireless devices are making an entry into the

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